# Asher Straus

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## EDUCATION

**Northwestern University** | McCormick School of Engineering | Evanston, IL — Sept 2023 – Mar 2028 (expected)
B.S. Manufacturing & Design Engineering | *Minor in Art* | *Segal Design Certificate* | *Robotics Certificate*
**GPA: 3.6 / 4.0 cumulative · 3.9 / 4.0 since sophomore year**

## WORK EXPERIENCE

**Siemens Healthineers** — *Mechanical Engineering Co-op, Molecular Imaging · Hoffman Estates, IL* — Jun 2026 – Present
- Updated CAD models and drawings for mechanical components in SPECT and PET scanners to resolve interferences and improve manufacturability. Applied GD&T to fully define part requirements.
- Designed fixtures and brackets in Siemens NX/Designcenter. Used 3D printing to evaluate and iterate on design.
- Contributed to the development of a multi-axis stage and investigated motorized drives to support SPECT research team.

**P1 Motor Club** — *Virtual Test Driver, Research & Promotions Intern · Tampa, FL* — Apr 2023 – Present
- Test new track layouts in simulators and give performance feedback to engineers to inform design changes.
- Promote the club and its track property to prospective members, including by building a large-scale, mostly 3D-printed model from the civil engineer's Civil 3D files, sized to travel in a truck as a membership sales tool.

**MIT Beaverworks Summer Institute** — *Teaching Assistant, "Basics of ASICs" · Cambridge, MA* — Jun 2024 – Aug 2024
- Guided students through ASIC schematic design, simulation, and testing, and helped refine labs and curriculum.

**Monticello Motor Club** — *Go-Kart Instructor, Technician & Social Media Coordinator · Monticello, NY* — Jun 2018 – Jul 2022
- Diagnosed and repaired kart systems for safe, reliable operation; coached drivers of all skill levels.

## RESEARCH & PROJECTS

**Advanced Intelligent Manufacturing Lab** — *Researcher · NSF ERC-HAMMER · MSEC 2026 Poster Co-author* — Nov 2025 – Apr 2026
- Smart Sheet Smith: co-author on the MSEC 2026 / NAMRC54 poster for a five-agent AI pipeline that turns a 2D sheet-metal drawing into a verified bending process plan, with no 3D CAD or labeled data.
- Built the Tracer Agent's evaluation corpus (CAD parts, multi-view drawings, hand-derived bend math) and a SolidWorks VBA pipeline that generates hundreds of valid drawings, cutting data creation from days to minutes.
- Fabricated 10+ flexible resistive tactile sensors (carbon nanotubes in Ecoflex) and ran near-field electrospinning at 10 kV (Fall 2025).

**Formula SAE Brake System** — *Suspension Member · Northwestern Formula Racing* — Sep 2024 – Present
- Hand-calculated the pedal force for a 1 G stop (~42 lbf pedal to ~690 lbf tire force), improved the team MATLAB brake sim, and verified rotor limits with ANSYS FEA.
- Machined and lathe-finished rotors, routed lines, and bled the system on the car.

**WaterGuard: CT Scanner for Aquatic Animals** — *Design & Build Engineer · Major Aquarium · Chicago, IL* — Sept 2025 – Mar 2026
- Designed a removable, reusable LDPE liner and runoff collector that waterproofs a ~$1M CT scanner used to image aquatic animals.
- Led liner geometry: an oblique frustum with a 12.5° runoff slope inside 6″ of table clearance, set by full-scale wrinkled-liner tests.
- Tensile-tested heat-welded seams: 94.8 N dry vs. 106.4 N after 24 h in saltwater, no weld separation; co-wrote the FMEA.
- On-site tests with veterinary staff cut setup from 45+ min to under 4; their laser-visibility feedback drove a clear PETG window.

**Performance Bicycle Brake Caliper** — *Simulation-Driven Design* — 2025
- Turned rider needs into specs (≤70 N input, ≤11 mm deflection, 15 m stop, ISO 4210); cut mass with FEA and topology optimization.
- SLS Nylon-12 prototype passed every metric on the bike: 41.3 g, 11.3 m stop at 68.9 N input, $8.26 per part.
- Led the stiffness–mass trade-off: a 14% lighter v2 needed 102 N at the lever, so we kept the first.

**Midwest EV** — *Mechanical Engineer · Tesla retrofit hardware and drive unit parts* — 2026 – Present
- Own the enclosure for the custom PCB in a Model S openpilot retrofit now sold at $2,280; five versions driven by weekly design reviews.
- Designing a replacement oil/coolant routing part for the Tesla large drive unit: converts rotor cooling from glycol to oil, so the rotor seal is lubricated and a leak can no longer rust the stator.

**Bionic Wrench Assembly Fixture** — *Designer & Manufacturing Engineer · DSGN 386* — Mar 2025 – Jun 2025
- Iterated 7 fixture versions; wrote the SOP, value stream map and line balance for a 1M-unit/year plan, with a pilot cycle time of 3:58.

**ExtendIt Desk Extension** — *Designer & Engineer · Human-Centered Product Design* — 2024
- Ran 22 user interviews and tested 7 mockups with 10 users each; 8 of 10 preferred the final locking desk extension (+50% surface).
- Built a load-test rig: held 33.5 lb at 3/8″ deflection against a 30 lb / 0.5″ target. Costed at $10.19 per unit for sale to universities.

**Pillowscape Sensory Pillow** — *Designer & Project Manager · Misericordia* — 2024 – Mar 2025
- Designed a customizable multi-texture sensory pillow for adults with developmental disabilities: swappable Velcro patches and a choice of filling, under $15 per unit against hundreds for commercial sensory devices.
- On-site testing with residents drove the huggable 20 × 30″ format; residents picked their own patches at handoff.

## TECHNICAL SKILLS

**Design & Fabrication:** Siemens NX/Designcenter, SolidWorks, Onshape, AutoCAD/Civil 3D, GD&T, Engineering Drawings, Fixture & Jig Design, 3D Printing, Laser Cutting, Waterjet, Machining (Lathe/CNC/Mill), Welding, Sand/Iron Casting, Sheet Metal, Metrology, Snap-Fit/Compliant Design, Rapid Prototyping, DFM/DFR

**Analysis & Software:** Ansys, FEA/Topology Opt., Thermal Sim., MATLAB, Python, VBA, SolidWorks API, LLM Pipelines, Embedded Systems/Raspberry Pi, FMEA, Cost Modeling, Lean/VSM, Line Balancing, Excel, Unity, GitHub, Photoshop, Circuit Design, Radio Control

**Clubs:** Formula SAE, Baja SAE, Combat Robotics, SME Student Chapter